



Eduardo G. Pacheco

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PROFILE

Data scientist and analytical chemist with 3+ years of experience in biotechnology R&D, machine learning, and scientific data analysis. Experienced in developing predictive models, automating laboratory workflows, and analyzing large biological datasets. Proven ability to optimize bioprocesses, improve product quality, and support clinical-scale manufacturing. Seeking data science or biotech R&D roles that leverage expertise in both experimental methods and computational tools.

SKILLS

- Python
- SQL
- Excel
- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Data Visualization
- Statistical Analysis
- Regression Analysis
- Classification
- Clustering
- Dimensionality Reduction
- Model Evaluation
- Model Tuning
- Git
- API Development
- AKTA FPLC
- HPLC
- TFF
- SDS-PAGE
- Western Blot
- ELISA
- DLS
- Chromatogram Analysis
- GMP
- GLP
- SOP Writing
- Tech Transfer
- Inventory Management
- Experimental Design
- Process Optimization

WORK EXPERIENCE

Process Development Associate I, Novavax Inc.

Nov 2024 - Apr 2025

- Built Python pipelines to automate chromatography data analysis and reporting, improving protein quantification accuracy and reducing manual documentation time by 40%.
- Developed a resin degradation monitoring framework, reducing column testing frequency and extending resin lifespan, improving both efficiency and cost savings.
- Led bench- and pilot-scale TFF purifications to evaluate seasonal COVID-19 and influenza vaccine strains, contributing to process scalability and performance validation.
- Conducted in-process analytical testing (SDS-PAGE, HPLC, Western Blot) and resolved product quality deviations observed at the pilot scale, supporting batch release decisions.
- Authored and delivered GMP-compliant tech transfer documents, presenting findings to cross-functional teams (QA, manufacturing, and analytical development) to guide downstream process improvements.

Research Associate, Vandstrom Inc.

Jun 2024 - Nov 2024

- Enhanced recombinant protein yield by refining AKTA FPLC protocols and automating the quantification of fluorescence-labeled
- Built regression models to optimize cell induction parameters, boosting expression by 20% and supporting experimental design improvements.
- Developed interactive dashboards in Python (Altair) to visualize purification trends and QC assay performance, enabling real-time insight for R&D scientists.
- Applied statistical methods, including ANOVA, regression analysis, and non-linear curve fitting, to support assay development, optimization, and validation.

Jul 2022 - Jun 2024

- Supported protein purification workflows using AKTA FPLC systems, ensuring consistent yield and quality in accordance with GMP standards.
- Maintained lab operations through buffer preparation, inventory management, and preventative maintenance, minimizing downtime and ensuring regulatory compliance.

Research Assistant, Virginia Commonwealth University - Richmond, VA

Jan 2020 - Mar 2022

- Conducted research on antiviral and anticancer metallodrugs and performed in vitro biological assays.
- Presented findings at department research expos and maintained detailed experimental records.

PROJECTS

Kaggle Competition: House Prices - Advanced Regression Techniques

Feb 2025 - Apr 2025

- Built and tuned multiple regression models (Ridge, Lasso, XGBoost) using Scikit-learn.
- Performed extensive feature engineering and log-transformations to improve performance.
- Achieved 0.14370 RMSE and deployed reproducible pipelines on GitHub.

EDUCATION

Master of Applied Data Science - 3.685

University of Michigan - Ann Arbor (In Progress)

Aug 2024 - May 2026

Bachelor of Science in Chemistry

Virginia Commonwealth University

Aug 2017 - May 2021

LANGUAGE

English: Fluent

Spanish: Fluent